Seminar: Artificial Intelligence - Visual Computing
3D Gaussian Splatting

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Why 3D Gaussian Splatting?

- **The NeRF Dilemma**
  Neural Radiance Fields (NeRFs) are either slow and provide photorealistic scenes or they are reasonably fast but have to sacrifice image quality. Real-time applications are not possible.

- **3D Gaussians As The Next Logical Step**
  Using 3D Gaussians as the explicit base primitive, cleverly decomposing them to make fast backpropagation possible, and abandoning neural networks is the key to unlocking previously unseen rendering speeds.

- **Engineering Effort**
  State-of-the-art GPU integration yields massive speed-ups, mostly due to the explicit nature of the base primitive.

Agenda

Core Concepts

3D Scene Creation

Novel-View Synthesis

Splatting

Definitions

Creates 3D environments such as individual objects or outdoor spaces from an overlapping set of 2D input images.

Core Concepts

3D Scene Creation

Novel-View Synthesis

Splatting

Definitions

Synthesizes new 2D imagery from the 3D scene using new camera positions.

Real-time capability requires at least 24–30 frames per second (FPS) to appear smooth enough for the human eye.

Core Concepts

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Novel-View Synthesis

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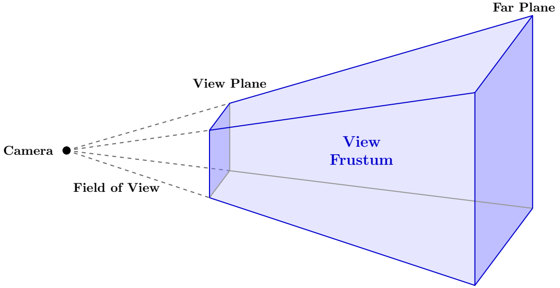
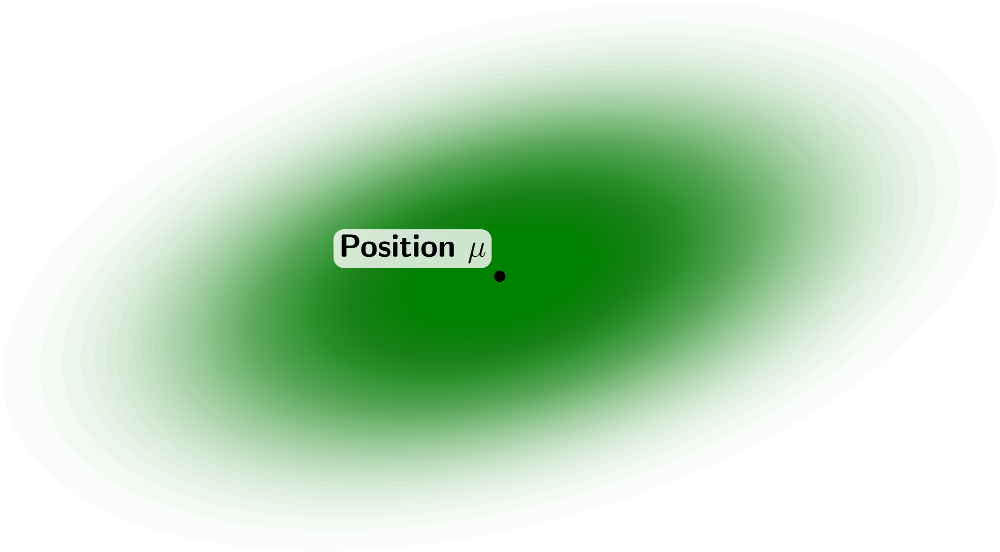
Definitions

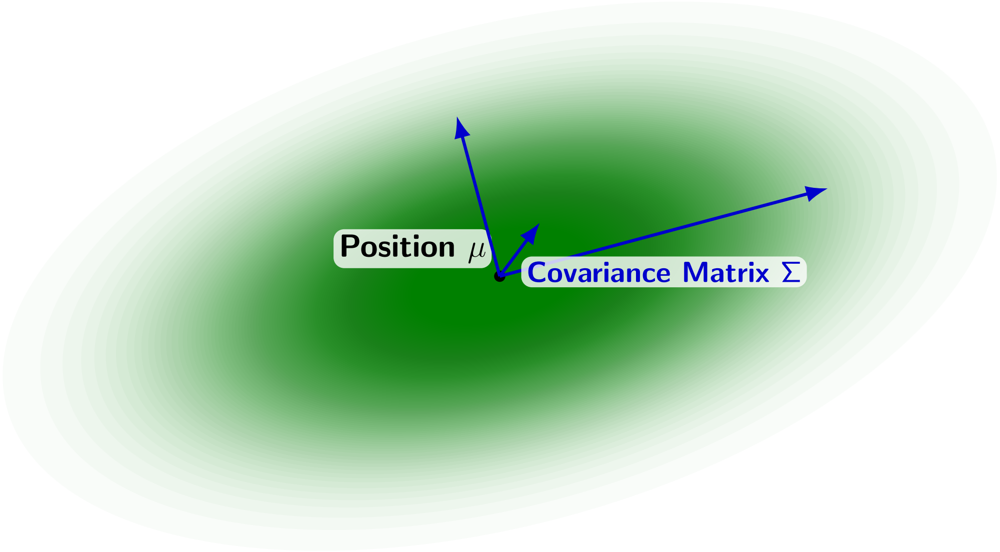
The process of moving all elements inside the view frustum onto the view plane preserving order, adding to aggregated color and opacity of each pixel.



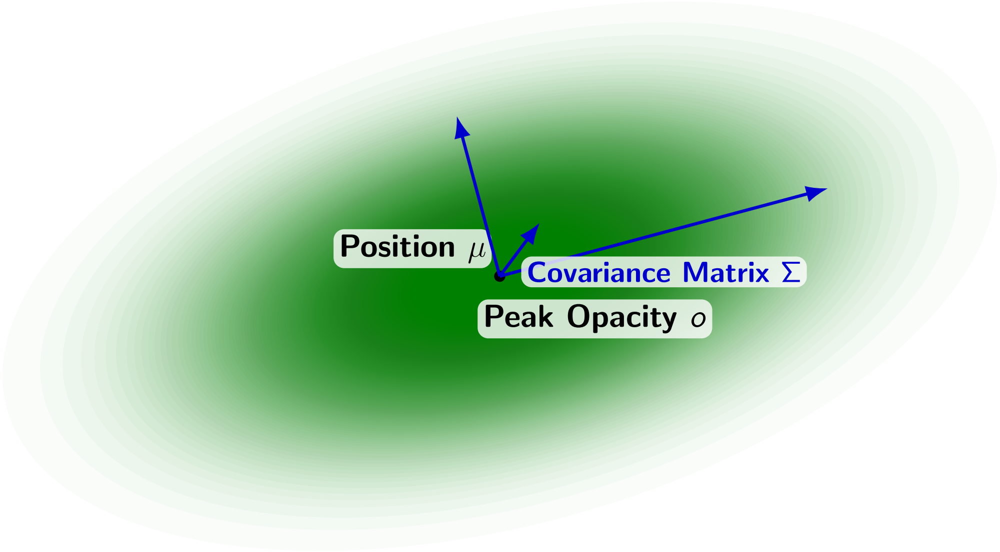
Overview

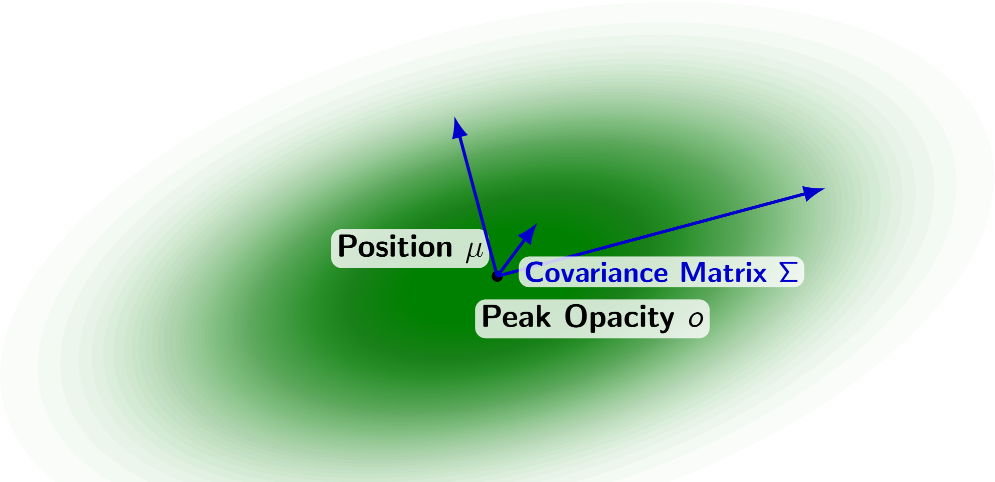
3D Gaussian Attributes


Position μ


Position μ
Covariance Matrix Σ

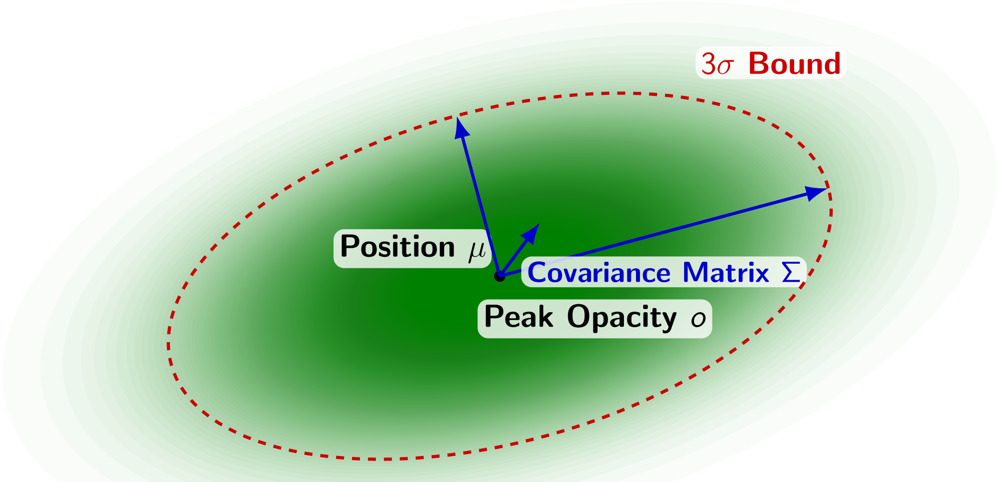
3D Gaussian Attributes





Multivariate Normal Distribution: $\alpha(p) = o \cdot exp\left(-\frac{1}{2}(p-\mu)^T\Sigma^{-1}(p-\mu)\right)$

3D Gaussian Attributes



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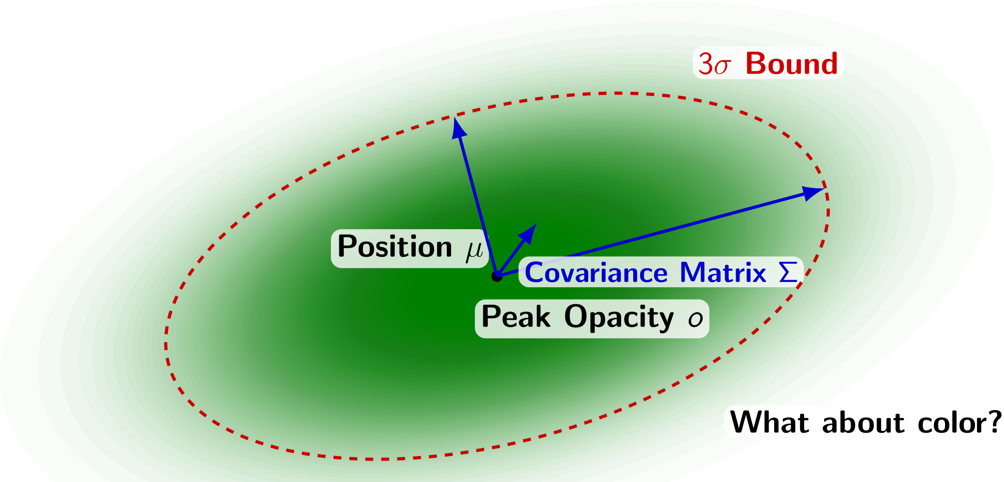

What about color?

Multivariate Normal Distribution: $\alpha(p) = o \cdot exp\left(-\frac{1}{2}(p-\mu)^T\Sigma^{-1}(p-\mu)\right)$

Mathematical Setup

The Validity Problem

Core Decomposition

Decomposed Components

Guaranteed Validity

Details

To work with gradient descent, the covariance matrix Σ must be differentiable and remain positive semi-definite (eigenvalues ≥ 0) to be geometrically valid.

Running a loop during backpropagation to find valid values is computationally too costly.

Mathematical Setup

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Details

Kerbl et al. bypassed this by decomposing Σ into a scaling matrix S and rotational matrix R mirroring standard eigendecomposition ($\Sigma = Q\Delta Q^T$):

$$\Sigma = RSS^TR^T \quad \text{with} \quad \Delta = S^2, Q = R$$

Mathematical Setup

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The diagonal scaling matrix S is further decomposed into three distinct scaling values. These represent the roots of the eigenvalues of Σ.

The rotational matrix R is decomposed into a four-value *quaternion*.

Mathematical Setup

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Decomposed Components

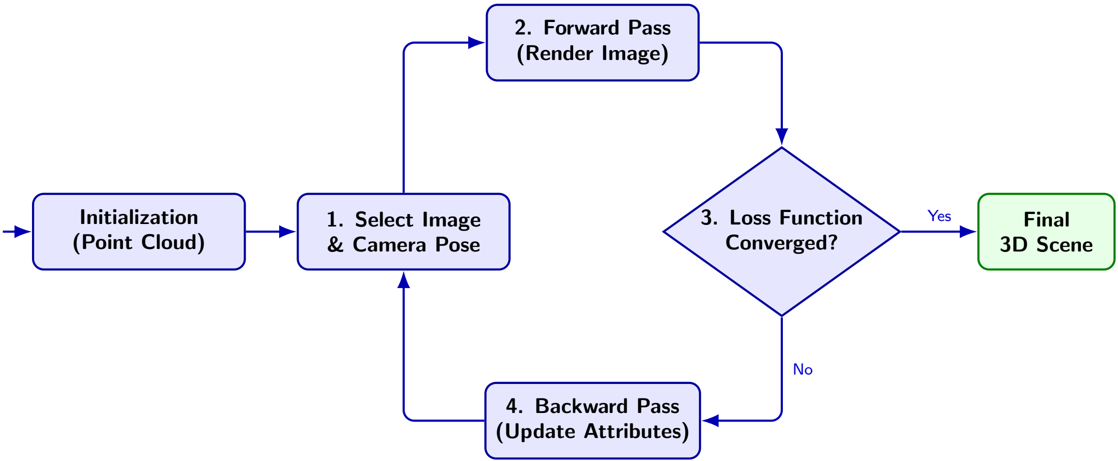
Guaranteed Validity

Details

3DGS only stores and optimizes these 7 specific values for each 3D Gaussian covariance matrix.

Because an eigendecomposition is always positive semi-definite, constructing a new covariance matrix from these components is guaranteed to be geometrically valid.

The Training Cycle



Overview

Initialization Steps

Structure from Motion

Camera Matrix

Primitive Initialization

Details

The 3DGS algorithm requires initial geometry, which is commonly derived from a set of input images.

Because camera poses are usually unknown, Structure from Motion (SfM) or similar algorithms are used to solve a geometric optimization problem, providing an accurate 3D point cloud.

Initialization Steps

Structure from Motion

Camera Matrix

Primitive Initialization

Details

As another result from SfM, each input image is assigned a camera matrix W which contains the camera's orientation in world space.

Initialization Steps

Structure from Motion

Camera Matrix

Primitive Initialization

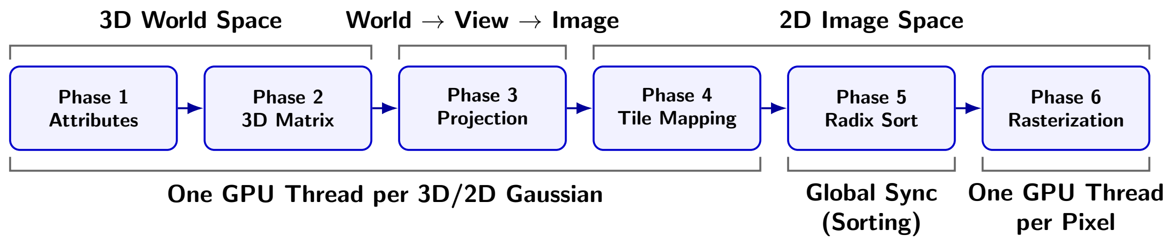
Details

The SfM points act as the origins for the initial 3D Gaussians.

- They are initialized as perfect spheres.
- Radii are deduced from their three nearest neighbors.
- Opacity is set to a low value, and color to any RGB value.

Overview

Forward Pass: Pipeline Architecture



Forward Pass: Phases 1 & 2

Phase 1: Preparing Constituent Attributes

- The GPU is running one GPU thread per 3D Gaussian.
- Raw values optimized during backpropagation are converted.
- Opacity is bounded to $(0, 1)$ using a sigmoid function, scale uses the exponential function, and the rotational quaternion is normalized.

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Phase 2: Reconstructing the 3D Covariance Matrix

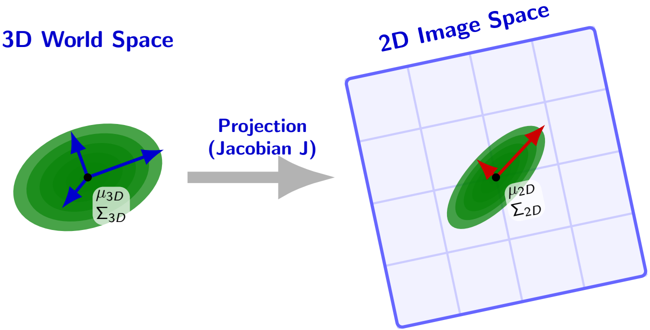
- The algorithm mathematically rebuilds the 3D geometric shape.
- It combines the scaling matrix S and rotational matrix R to form the full 3D covariance matrix:

$$\Sigma_{3D} = RSS^T R^T$$

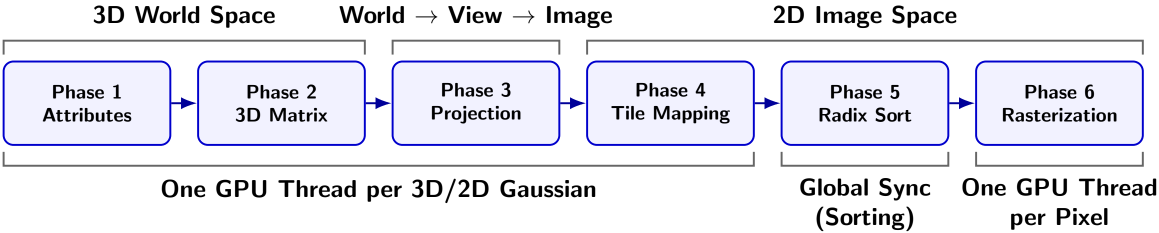
Phase 3: Projection from 3D to 2D

- Converts 3D Gaussians (ellipsoids) in world space into flat 2D Gaussians (ellipses) in image space.
- Uses a linear affine approximation to keep the 2D Gaussian elliptic (Jacobian J).
- The 3D matrix is rotated into view space via camera matrix W and then projected:

$$\Sigma_{2D} = JW\Sigma_{3D}W^TJ^T$$

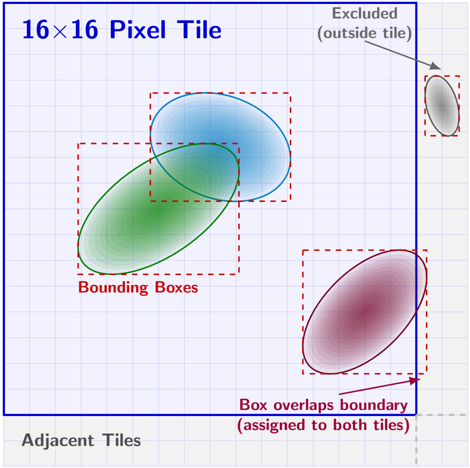


Forward Pass: Pipeline Architecture



Phase 4: Tile Mapping (Preparing Radix Sort)

- The target output image is divided into a grid of tiles of 16×16 pixels.
- The algorithm calculates a 2D bounding box to map exactly which tiles a 2D Gaussian covers.
- Assigns each 2D Gaussian to a tile. If overlapping, assigns to all affected tiles.



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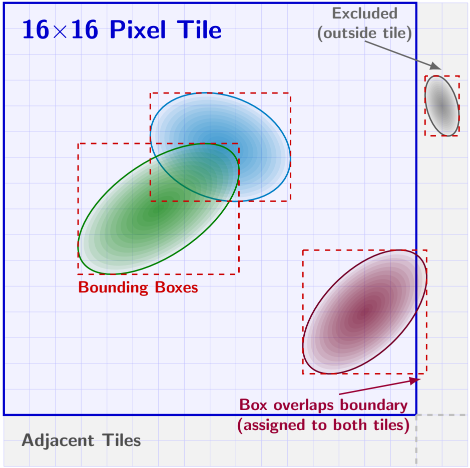
Phase 5: Global Radix Sort

- The GPU waits until all per-Gaussian computations are finished.
- A highly optimized Radix Sort is executed, organizing the 2D Gaussians for each tile strictly by depth (closest to farthest).



Phase 6: Tile-Based Rasterization (Splatting)

- Operating in 2D image space, the GPU assigns one processing thread per pixel.
- **Alpha Compositing:** Pixels accumulate color by blending the sorted list of overlapping 2D Gaussians front-to-back:

$$C_p = \sum_{i=1}^{N} c_i \alpha_i T_i, \quad \text{with} \quad T_i = \prod_{j=1}^{i-1} (1 - \alpha_j) \quad \text{the transmittance}$$

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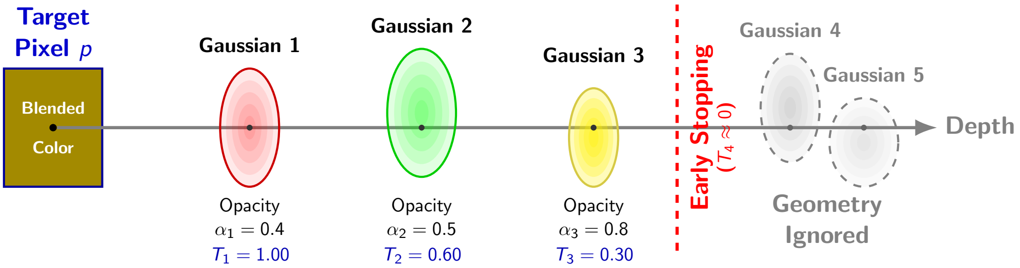
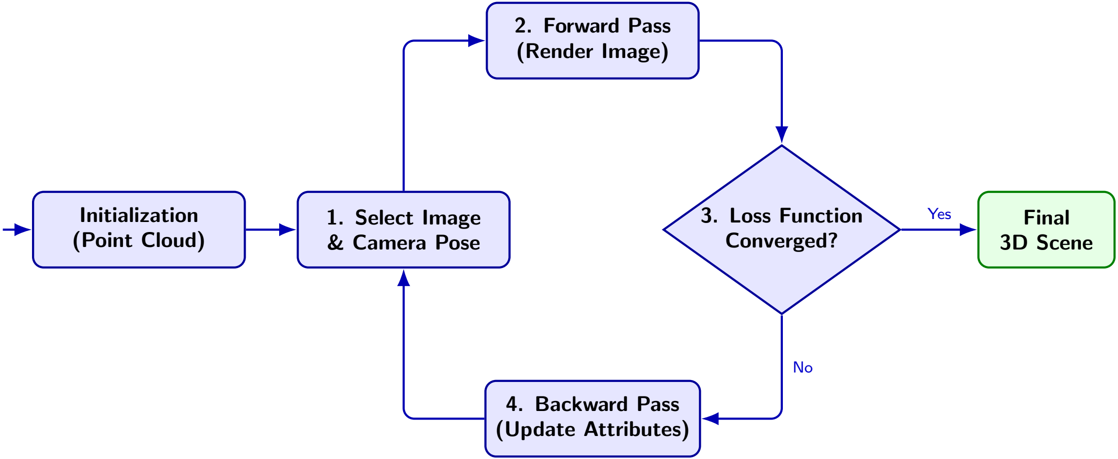
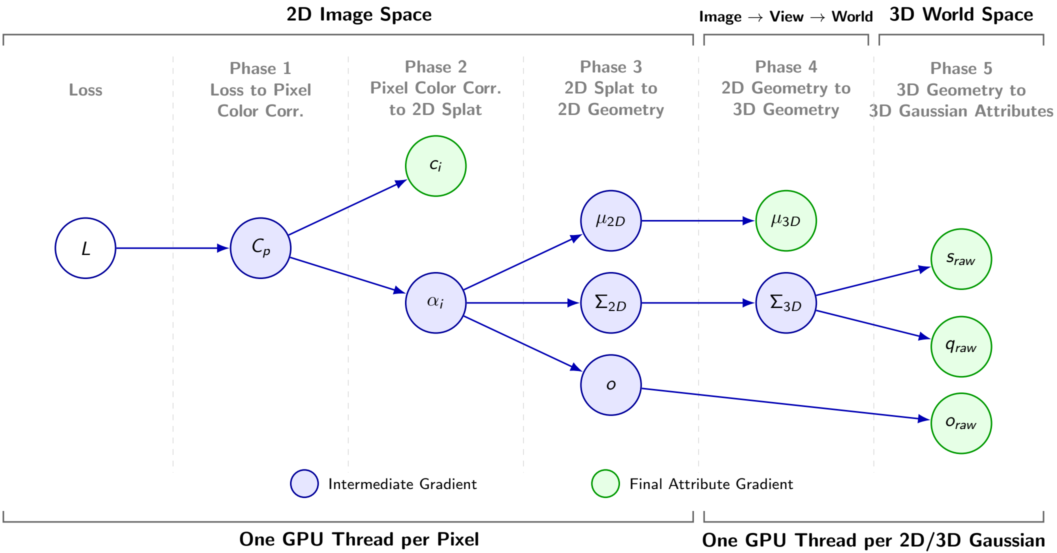


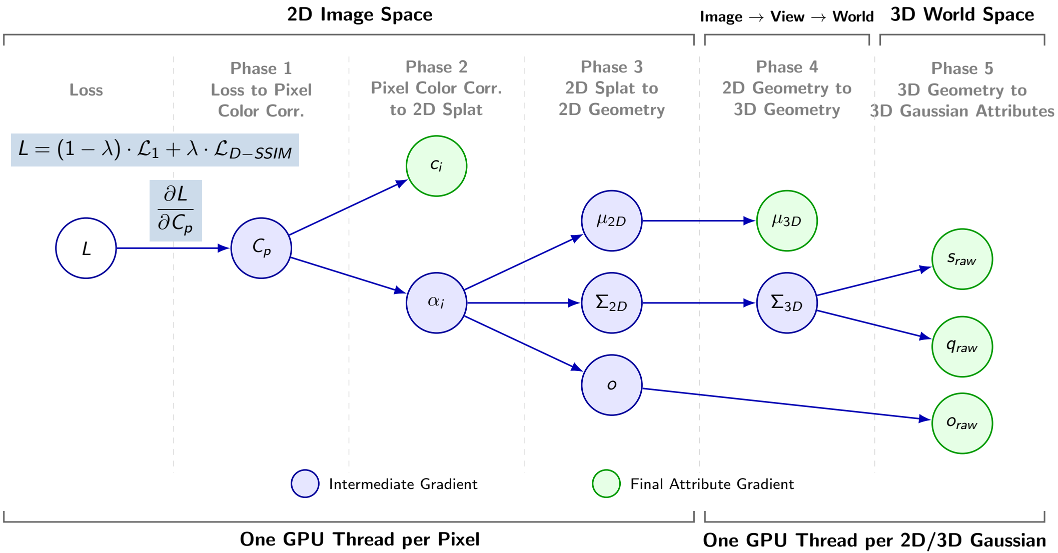
The Training Cycle



Overview

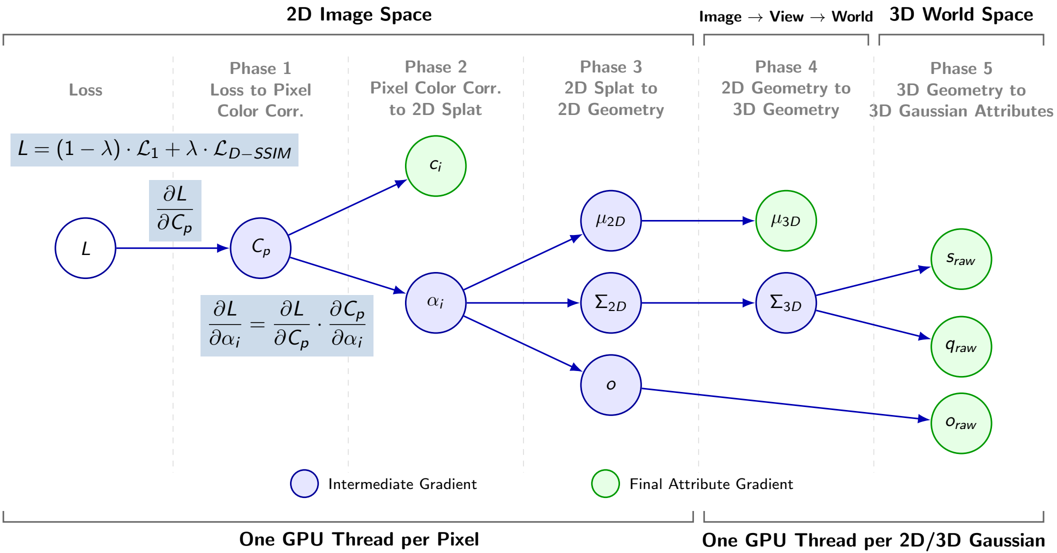
Backward Pass: Architecture & Gradient Flow





2D Image Space
Image → View → World
3D World Space
Loss
Phase 1
Loss to Pixel
Color Corr.
Phase 2
Pixel Color Corr.
to 2D Splat
Phase 3
2D Splat to
2D Geometry
Phase 4
2D Geometry to
3D Geometry
Phase 5
3D Geometry to
3D Gaussian Attributes
$L = (1 - \lambda) \cdot \mathcal{L}_1 + \lambda \cdot \mathcal{L}_{D-SSIM}$
$\frac{\partial L}{\partial C_p}$
L
C_p
c_i
α_i
μ_{2D}
Σ_{2D}
o
μ_{3D}
Σ_{3D}
s_{raw}
q_{raw}
o_{raw}
Intermediate Gradient
Final Attribute Gradient
One GPU Thread per Pixel
One GPU Thread per 2D/3D Gaussian

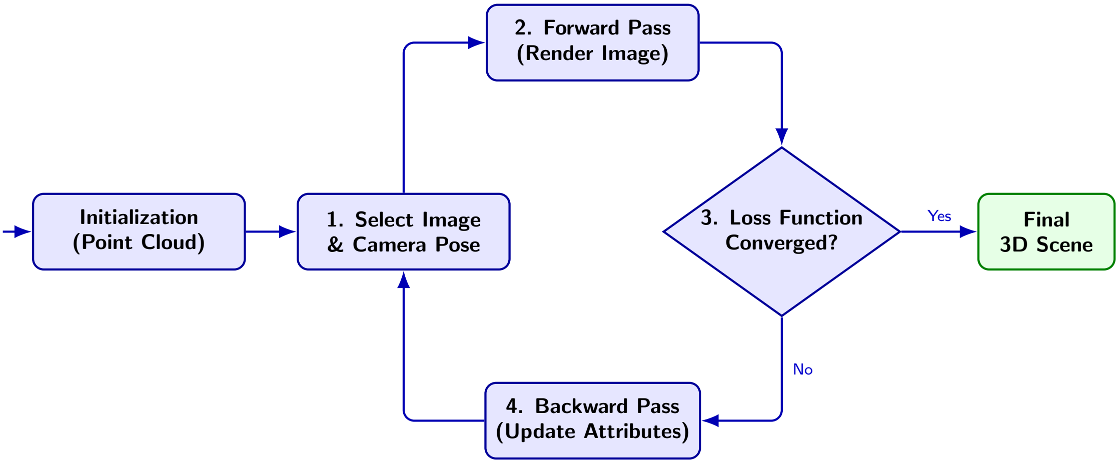
Backward Pass: Architecture & Gradient Flow



Adaptive Density Control

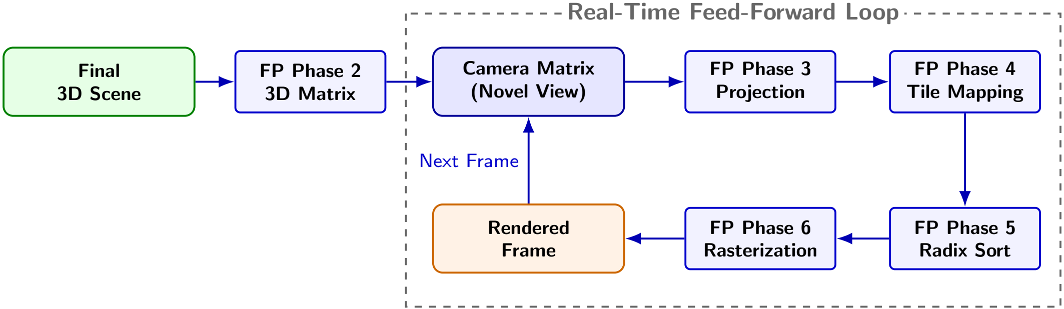
- Steps in every 100 iterations to counter over-reconstruction (covering too much geometry) and under-reconstruction (too few Gaussians).
- **Splitting:** Over-reconstructing Gaussians are split into two smaller ones.
- **Cloning:** Under-reconstructing Gaussians are cloned.

The Training Cycle



Overview

Novel-View Synthesis



- Because the scene geometry is finalized, the backward pass, Adaptive Density Control, and other components are completely disabled.
- Freed from the this overhead and running at the raw speed of the tile-based rasterizer, 3DGS quickly reaches real-time rendering levels.

Overview

Summary and Discussion

- 3D Gaussian Splatting is a learnable and explicit radiance field representation for 3D Reconstruction, abandoning neural networks.
- Kerbl et al. combined multiple existing approaches and technologies, added new ideas, and mapped them onto modern GPU hardware.
- We analyzed the training procedure showing in detail how Kerbl et al. made it possible to have real-time rendering of reconstructed 3D scenes for the first time.

Thank you for your attention! I am looking forward to taking your questions.

Overview

3DGS Videos

Training in Progress

https://youtu.be/F9rgbvhSad8?t=550

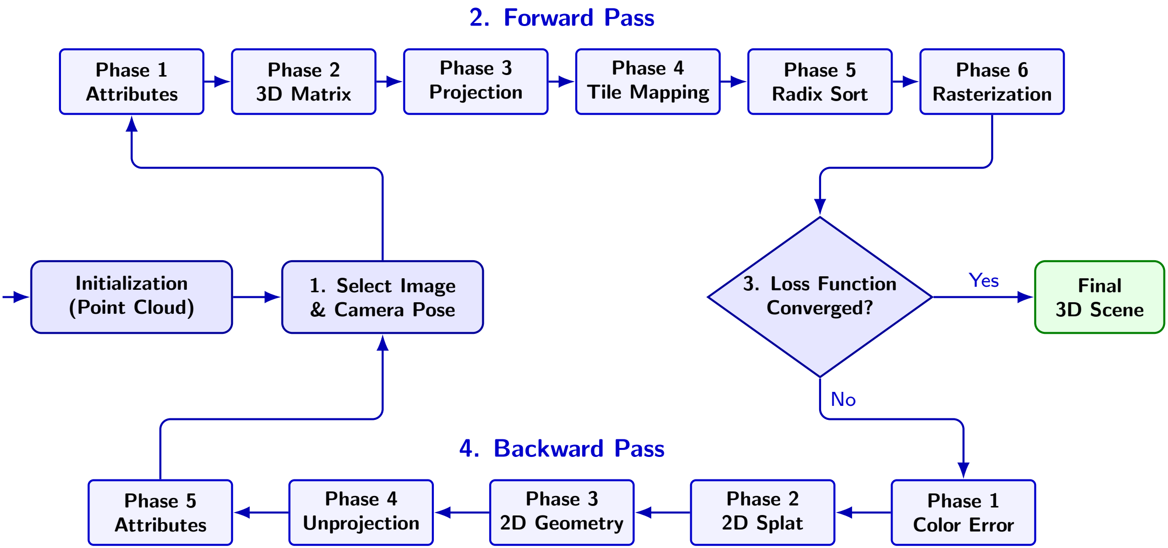
https://youtu.be/yd40zUnr7eQ

Real-Time Rendering

Example from the original paper's project website

https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/content/videos/bicycle.mp4

The Detailed Training Cycle



Overview

The Loss Function

- Training compares the rendered image with the original input image using a weighted combination of $\mathcal{L}_1$ loss and D-SSIM.
- $\mathcal{L}_1$ calculates the absolute pixel-wise error, while D-SSIM acts as a structural regularizer:

$$L = (1 - \lambda) \cdot \mathcal{L}_1 + \lambda \cdot \mathcal{L}_{D-SSIM}, \quad \text{with } \lambda = 0.2$$

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Phase 1: Loss to Pixel Color Correction

- The gradient calculates the required color correction per pixel (C_p).
- A positive gradient indicates an overestimation of color, negative indicates underestimation, and the magnitude dictates the degree of adjustment required:

$$\frac{\partial L}{\partial C_p} = \frac{1 - \lambda}{N} \cdot sign(C_p - O_p) + \lambda \frac{\partial \mathcal{L}_{D-SSIM}}{\partial C_p}$$

Phase 2: Pixel Color Correction to 2D Splat

- The algorithm works alpha compositing backwards, flowing the pixel gradient into the evaluated opacity (α_i) and color (c_i) of each 2D Gaussian in the tile.
- **Color Gradient:** Uses the chain rule to update the Gaussian's color based on its contribution (transmittance T_i and opacity α_i):

$$\text{with } C_p = \sum_{i=1}^{N} c_i \alpha_i T_i : \quad \boxed{\frac{\partial L}{\partial c_i} = \frac{\partial L}{\partial C_p} \cdot \frac{\partial C_p}{\partial c_i} = \frac{\partial L}{\partial C_p} \cdot (\alpha_i T_i)}$$

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- **Opacity Gradient:** Increases evaluated opacity if the Gaussian's color (c_i) reduces the pixel error more effectively than the accumulated background color behind it (C_{behind}):

$$\boxed{\frac{\partial L}{\partial \alpha_i} = \frac{\partial L}{\partial C_p} \cdot \frac{\partial C_p}{\partial \alpha_i} = \frac{\partial L}{\partial C_p} \cdot T_i(c_i - C_{behind})}$$

Phase 3: 2D Splat to 2D Geometry

- The opacity gradients are used to adjust the physical properties of the 2D Gaussians.
- This mathematically pulls the center position of the 2D Gaussian towards the pixel and stretches or contracts its 2D shape to minimize the error.

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Phase 4: 2D Geometry to 3D Geometry

- The GPU model changes from pixel-based to running per-Gaussian calculations.
- The 2D gradients are unprojected back into 3D space, translating the 2D adjustments into 3D center movements and 3D volume changes.

Backward Pass: Phases 3 to 5

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Phase 5: 3D Geometry to 3D Gaussian Attributes

- The gradients flow backward through the 3D covariance matrix assembly.
- Adjustments are mapped directly into the raw variables for scale, rotation, and base opacity.

Overview

Phase 3: 2D Splat to 2D Geometry (Opacity & Position)

- We recall the simplified opacity distribution: $\alpha_i(p) = o \cdot e^E$, where E is the exponent derived from the pixel distance Δ and the inverse 2D covariance matrix.
- **Base Opacity (o):** The derivative scales directly with the distance from the center of the 2D Gaussian (e^E):

$$\frac{\partial L}{\partial o} = \frac{\partial L}{\partial \alpha_i} \cdot e^E$$

- **Position (μ_{2D}):** Let vector $v = \Sigma_{2D}^{-1} \Delta p$. This vector points directly from the 2D Gaussian's center to the pixel:

$$\frac{\partial L}{\partial \mu_{2D}} = \frac{\partial L}{\partial \alpha_i} \cdot \alpha_i(p) \cdot v$$

- If the evaluated opacity gradient is positive, the center is effectively pulled towards the pixel.

Phase 3: 2D Splat to 2D Geometry (Shape)

- **2D Covariance Matrix (Σ_{2D}):** We must calculate how the opacity at pixel p changes when the 2D shape of the Gaussian changes.
- This uses the same vector v from the position derivative, multiplied by its own transpose:
$$\frac{\partial L}{\partial \Sigma_{2D}} = \frac{\partial L}{\partial \alpha_i} \cdot \frac{1}{2}\alpha_i(p) \cdot (v \cdot v^T)$$
- Multiplying the vector by its transpose results in a matrix gradient.
- If the evaluated opacity gradient is positive, this matrix gradient causes the 2D Gaussian to stretch itself towards the pixel; otherwise, it causes it to contract.

Phase 4: 2D Geometry to 3D Geometry

- The GPU thread model changes to running per-Gaussian to unproject the 2D data back into 3D.
- This utilizes the multi-variable chain rule and the transposes of the forward projection matrices ($U = JW_{rot}$).
- **3D Position (μ_{3D}):** The gradient passes through view space (t) back to world space:

$$\frac{\partial L}{\partial \mu_{3D}} = W_{rot}^T J^T \frac{\partial L}{\partial \mu_{2D}} = U^T \left(\frac{\partial L}{\partial \mu_{2D}} \right)$$

- **3D Covariance (Σ_{3D}):** The final gradient unprojects cleanly using matrix U on both sides:

$$\frac{\partial L}{\partial \Sigma_{3D}} = U^T \left(\frac{\partial L}{\partial \Sigma_{2D}} \right) U$$

Phase 5: 3D Geometry to Matrix Components

- Gradients flow through the covariance matrix assembly ($\Sigma_{3D} = MM^T$ with $M = RS$) into scale (S) and rotation (R).
- First, the algorithm derives the intermediate matrix M:

$$\frac{\partial L}{\partial M} = 2\left(\frac{\partial L}{\partial \Sigma_{3D}}\right) M$$

- Then, it splits this gradient into the separate components:

$$\frac{\partial L}{\partial S} = R^T\left(\frac{\partial L}{\partial M}\right) \quad \text{and} \quad \frac{\partial L}{\partial R} = \left(\frac{\partial L}{\partial M}\right) S$$

- The scale gradient ($\frac{\partial L}{\partial s}$) is extracted from the diagonal of $\frac{\partial L}{\partial S}$, while the quaternion gradient ($\frac{\partial L}{\partial q}$) aggregates the partial derivatives of R.

Phase 5: Backpropagating to Latent Attributes

- To close the loop, the gradients pass backwards through their respective activation functions to reach the raw, latent space variables.
- **Scale (Exponential):** Backpropagating through exp requires multiplying by s:

$$\frac{\partial L}{\partial s_{raw}} = \frac{\partial L}{\partial s} \cdot s$$

- **Rotation (Normalization):** Uses the Jacobian matrix ($J_{||}$) of the normalization function, with $J_{||} = \frac{I - qq^T}{||q_{raw}||}$:

$$\frac{\partial L}{\partial q_{raw}} = J_{||} \cdot \frac{\partial L}{\partial q}$$

- **Base Opacity (Sigmoid):** Backpropagating through the sigmoid function:

$$\frac{\partial L}{\partial o_{raw}} = \frac{\partial L}{\partial o} \cdot o(1 - o)$$

Overview

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